• Monsoon, Forest, Soil, Irrigation Projects, Agriculture

21. Why does the Government of India promote the use of 'Neem -coated Urea' in agriculture ?

(a) Release of Neem oil in the soil increases nitrogen fixation by the soil microorganisms.

(b) Neem coating slows down the rate of dissolution of urea in the soil

(c) Nitrous oxide, which is a greenhouse gas, is not at all released into the atmosphere by crop fields.

(d) It is a combination of a weedicide and a fertilizer for particular crops

Ans. (b): Urea, which is necessary for the development of plants, is an important supplier of nitrogen. But only 30-40 percent of nitrogen present in the area is utilized by crops and the rest gets degraded in various stages.

• When ordinary urea is applied, it gets converted to ammonium carbonate. Some of this gets converted to ammonia gas in what is called ammonia volatilisation.

•About 8-10 percent of nitrogen is lost during volatilisation.

• Neem has properties that check nitrogen loss at each stage. It slows down the process of nitrate formation and hence excess nitrate is not available for desertification. Hence, Neem coating slows down the rate of dissolution of urea in the soil.

• Minerals Energy Resources, Industry and Trade

22. What is /are the purpose /purposes of 'District Mineral Foundations' in India ?

1. Promoting mineral exploration activates in mineral-rich districts
2. Protecting the interests of the persons affected by mining operations
3. Authorizing State Governments to issue licenses for mineral exploration .

Select the correct answer using the code given below :

(a) 1 and 2 only (b) 2 only

(c) 1 and 3 only (d) 1, 2 and 3

Ans. (b): District Mineral Foundation (DMF) is a trust set up as a non-profit body in those districts affected by the mining works to work for the interest and benefit of persons and areas affected by mining -related operations. It is founded through the contributions from miners. Its manner of operation comes under the jurisdiction of the relevant State Government.

23. In which of the following regions of India are shale gas resources found ?

1. Cambay Basin
2. Cauvery Basin
3. Krishna-Godavari Basin

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 3 only

(c) 2 and 3 only (d) 1, 2 and 3

Ans. (d): Shale gas is natural gas that is found trapped within non-porous rocks.

• India has identified 6 basins as areas for shale gas exploration : Cambay (Gujrat), Assam-Arakan (North-East), Gondwana (Central India), Krishna Godavari onshore and Indo-Gangetic basins.

• Transport

24. Recently, which of the following States has explored the possibility of constructing an artificial inland port to be connected to sea by a long navigational channel ?

(a) Andhra Pradesh (b) Chhattisgarh

(c) Karnataka (d) Rajasthan

Ans. (d): The government of Rajasthan is planning to develop an Inland shipping port at Jalore, which would be connected to the Arabian Sea by developing a channel along the Kutch Creek.

World Geography

• Atmosphere

25. In the cities of our country, which among the following atmospheric gases are normally considered in calculating the value of Air Quality Index ?

1. Carbon dioxide
2. Carbon monoxide
3. Nitrogen dioxide
4. Sulfur dioxide
5. Methane

Select the correct answer using the code given below :

(a) 1, 2 and 3 only (b) 2, 3 and 4 only

(c) 1, 4 and 5 only (d) 1, 2, 3, 4 and 5

Ans. (b): National Air Quality Index : There are 6 AQI categories, namely Good, Satisfactory, Moderately polluted, Poor, Very, Poor and Severe. The AQI considers eight pollutants (PM 10 , PM 2.5, NO_2, SO_2, CO, O_3, NH_3 and Pb) for which short-term (upto 24-hourly averaging period). National Ambient Air Quality Standards are prescribed.

• Human Geography and Mapping

26. Consider the following pairs :

Community sometimes mentioned in the news		In the affairs of
1. Kurd	–	Bangladesh
2. Madhesi	–	Nepal
3. Rohingya	–	Myanmar

Which of the pairs given above is/are correctly matched ?

(a) 1 and 2 (b) 2 only

(c) 2 and 3 (d) 3 only

Ans. (c): Kurds are an ethnic group in the Middle East, so option (1) is incorrect.

Madhesis were in the news due to Nepal's constitutional debates about their representation in the polity. So (2) is correct. Rohingyas Muslims and Buddhist rakhines have been indulging into fights and riots in Myanmar, which has frequently been in news.

So, (3) is correct.

Indian Constitution and Governance

• Legislature : Council of Ministers

27. The Parliament of India acquires the power to legislate on any item in the State List in the national interest if a resolution to that effect is passed by the –

(a) Lok Sabha by a simple majority of its total membership

(b) Lok Sabha by a majority of not less then two-thirds of its total membership

(c) Rajya Sabha by a simple majority of its total membership

(d) Rajya Sabha by a majority of not less than two - thirds of its members present and voting

Ans. (d) : If Rajya Sabha passes a resolution by a majority of not less than two-thirds of numbers present and voting saying it is "necessary or expedient in the "national interest" that Parliament should make a law on a matter enumerated in the State List, Parliament becomes empowered to make a law on the subject specified in the resolution, for the whole as any parts of the territory of India. Such as resolution remains in force for a maximum period of one year but this period can be extended by one year at a time by passing a similar resolution further.

28. Which of the following statements is/are correct

1. A Bill pending in the Lok Sabha lapses on its prorogation.
2. A Bill pending in the Rajya Sabha, which has not been passed by the Lok Sabha, shall not lapse on dissolution of the Lok Sabha.

Select the correct answer using the code given below

(a) 1 only

(b) 2 only

(c) both 1 and 2

(d) Neither 1 nor 2

Ans. (b): When the Lok Sabha is dissolved, all business, including bills, motions, resolutions, notices, petitions and so on pending before it or its committees lapse. They need to be reintroduced in the newly constituted Lok Sabha.

A bill pending in the Rajya Sabha but not passed by the Lok Sabha does not lapse. A bill passed by the Lok Sabha but pending in the Rajya Sabha lapses. A bill passed by both Houses but pending assent of the president does not lapse.

• Centre-State Relations, Administration, Jammu and Kashmir

29. Consider the following statements :

1. The Chief Secretary in a State is appointed by the Governor of that State.
2. The Chief Secretary in a State has a fixed tenure

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d) : The Incumbent to the post of chief secretary is chosen by and appointed by the chief minister and ministers of the state, the trend is that the CM consults the union government regarding the appointments of the chief secretary but the consultation is not obligatory.

• Judicial System (Centre and State)

30. With reference to the 'Gram Nyayalaya Act' which of the following statements is/are correct ?

1. As per the Act, Gram Nyayalayas can hear only civil cases and not criminal cases
2. The Act allows local social activists as mediators/ reconciliators.

Select the correct answer using the code given below–

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b): The Gram Nyayalayas are mobile village courts in India established under Gram Nyayalayas Act -2008 for speedy and easy access to the justice system in the rural area. The Yojana mainly addresses the longevity risks among the workers in unorganised sector and encourages them to voluntarily save for their retirement. So, (1) is correct.